

Job Aggregation Platform - Final Project Report

Reuben Jacob

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Job Aggregation Platform — Final Project Report

Internal Engineering Report · Data Platform

Prepared for	Engineering Manager
Prepared by	Reuben Jacob
Report date	July 15, 2026
Data snapshot as of	July 10, 2026
Companion document	<code>Final_Project_Report_v2.md</code> (full technical detail, methodology, and evidence trail)

This is a condensed version of `Final_Project_Report_v2.md`, written for a single reading in approximately 15–20 minutes. No technical finding has been removed — long narrative explanations have been replaced with tables, and repeated explanations have been cut. All figures are sourced from this project's PostgreSQL database, source code, and `job_api.xlsx` research workbook (API Catalog and Sheet1). Where a fact could not be verified from these sources, this report states **"Not publicly available"** or **"Not verified during this project"** rather than estimating.

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1. Project Overview

Purpose. Build a single, de-duplicated, queryable dataset of job postings, sourced from as many legitimately accessible providers as possible, normalized into one schema with country and remote/hybrid/on-site classification, and persisted in a real database rather than a flat file.

Scope. The project researched **268 job-related APIs, ATS platforms, and job boards** (government labour agencies, commercial aggregators, ATS platforms, and job boards), narrowed that list to an **86-provider shortlist** for close audit, and built **46 working integrations** against a shared `sources/` interface (`BaseJobSource`). Every researched provider — implemented or not — is documented with a specific, evidence-based reason (Section 4).

Final outcome.

- **46 providers implemented**; **41 currently contributing live data** to PostgreSQL.
- **184,506 unique job records** stored, deduplicated by URL/hash, with per-provider run history.
- **Offline country and work-arrangement normalization** — no paid geocoding service used.
- The remaining **222 researched providers** were not implemented for specific, documented reasons: no public API, enterprise/partner-only access, wrong data-flow direction (employer-posting tools, not read APIs), deprecated services, or a registration/verification step not yet completed.
- The platform reflects **one complete backfill run**, not yet a recurring schedule — it is ready for operational hardening, not further architecture, before production use.

2. Project Status

Area	Status	Detail
Provider research (268 researched, 86 shortlisted)	Completed	Every provider carries a documented reason for its outcome (Section 4)
46 provider integrations (shared <code>BaseJobSource</code> interface)	Completed	One module per provider under <code>sources/</code>
PostgreSQL persistence (<code>jobs</code> , <code>provider_runs</code> tables)	Completed	Upsert-on- <code>dedup_key</code> , idempotent schema
Deduplication (URL/hash-based)	Completed	0 cross-provider duplicate URLs remaining
Offline country normalization	Completed	Multi-country/region text forced to <code>NULL</code> rather than guessed
Work-arrangement classification (remote/hybrid/on-site)	Completed	Derived from location text and provider-native flags
<code>jobs.json</code> static export + frontend search/filter	Completed	Unchanged since Milestone 2
Per-source fault isolation	Completed	One broken provider cannot halt a run
41 of 46 providers contributing live data	Partially Completed	5 non-contributing for external reasons (billing, missing credentials, one zero-result run) — Section 3
Employment-type capture	Partially Completed	Present for 24 of 46 providers; absent for the rest
Country resolution long tail	Partially Completed	Multi-country case solved; ~1,565 distinct unresolved single-place values remain
Credentialed-provider setup	Partially Completed	2 of 12 credentialed providers (Trade Me, CareerOneStop) lack credentials in this environment
6 real, accessible candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)	Partially Completed	Registration/verification step outstanding, not rejected
Standardized job-category taxonomy	Not Started	No cross-provider taxonomy exists in the data (Section 5, item 11)
Salary / visa-sponsorship capture	Not Started	No schema field; no provider exposes it in a usable form
Scheduled/automated recurring syncs	Not Started	Every run to date has been a single manual invocation
Inline normalization at insert time	Not Started	Currently a separate, manually-triggered migration script
Provider-run health alerting	Not Started	No automated flag for a source dropping to 0 jobs
Dashboard/analytics UI	Not Started	Static search frontend only
Expired-job / posting-removal detection	Not Started	Requires a second ingestion run to diff against; none has occurred

3. Implemented APIs

3.1 LIFECYCLE SUMMARY

Stage	Count
Providers researched (API Catalog sheet, <code>job_api.xlsx</code>)	268
Providers shortlisted for close audit (Sheet1, <code>job_api.xlsx</code>)	86
Providers implemented (<code>sources/*.py</code> , registered in <code>sources/__init__.py</code>)	46
Providers currently contributing live data	41
Providers implemented but not currently contributing	5
Providers researched but not implemented	222

3.2 WHY 5 IMPLEMENTED PROVIDERS ARE NOT CURRENTLY LIVE

Provider	Reason
TheirStack	Account's billing plan does not include Jobs API access (HTTP 402)
Trade Me Jobs	Built, but no production credentials configured; also requires Trade Me's approval process
CareerOneStop	Built, but no credentials configured (<code>CAREERONESTOP_USER_ID</code> / <code>API_TOKEN</code>)
Fantastic.jobs	Credentialed, but returned 0 jobs in the latest run — most plausibly a trial-window/account-tier issue, not confirmed
Arbeitsnow	Free, no-auth API that previously worked (931 jobs in earlier testing); returned 0 jobs in the latest run — cause not confirmed

3.3 VOLUME CONCENTRATION

Provider	Jobs Stored	% of Total
SmartRecruiters	65,392	35.4%
Greenhouse	22,715	12.3%
Lever	17,980	9.7%
NHS Jobs	10,843	5.9%
USAJOBS	10,000	5.4%
Bundesagentur für Arbeit	9,896	5.4%
Reed	8,994	4.9%
NAV Arbejdsplassen	5,394	2.9%
Ashby	5,252	2.8%
Workable	4,137	2.2%

Top 3 providers supply 57.4% of all 184,506 stored jobs — volume is concentrated in ATS boards, not evenly spread across all 46 sources.

4. Complete Provider Research (268 Providers)

Every provider researched in this project (`job_api.xlsx` API Catalog sheet, 268 rows), split into **Implemented (46)** and **Not Implemented (222)**. Pricing model and payment terms are this project's own verified classification of each provider's published terms — where nothing could be confirmed, the cell reads "Not publicly available." Reference is a documented URL/domain found in this project's research notes.

4.1 IMPLEMENTED PROVIDERS (46)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
SmartRecruiters	No-auth public API	No	developers.smartrecruiters.com	65,392 stored (65,394 fetched)	Live	• Limitation: 100 curated companies only, not a global index. • Expansion: Add more verified company slugs.
Greenhouse	No-auth public API	No	developers.greenhouse.io	22,715 stored (22,715 fetched)	Live	• Limitation: 276 curated companies; single-shot fetch per company. • Expansion: Add more verified company slugs.
Lever	No-auth public API	No	github.com	17,980 stored (17,980 fetched)	Live	• Limitation: 162 curated companies; single-shot fetch per company. • Expansion: Add more verified company slugs.
NHS Jobs	No-auth public feed	No	jobs.nhs.uk	10,843 stored (13,011 fetched)	Live	• Limitation: Undocumented endpoint; could change without notice. • Expansion: Monitor for endpoint changes; no code change needed.
USAJOBS	Free registration	No	developer.usajobs.gov	10,000 stored (10,000 fetched)	Live	• Limitation: Hard 10,000-result API ceiling; no retry/error handling in fetch loop. • Expansion: Partition by JobCategoryCode/LocationName; add shared retry helper.
Bundesagentur für Arbeit	No-auth-style public key	No	github.com	9,896 stored (10,000 fetched)	Live	• Limitation: Hard 10,000-result API ceiling (page x size). • Expansion: Partition query by location/keyword params.
Reed	Free registration	No	reed.co.uk	8,994 stored (9,000 fetched)	Live	• Limitation: ~9,900 result-window boundary (HTTP 500 beyond); no shared retry helper. • Expansion: Raise page cap toward boundary; add shared retry helper.
NAV Arbejdsplassen	No-auth-style public token	No	navikt.github.io	5,394 stored (12,478 fetched)	Live	• Limitation: Change-feed, not a current-postings search; drives high dup rate. • Expansion: Increase lookback window or page cap.
Ashby	No-auth public API	No	developers.ashbyhq.com	5,252 stored (5,252 fetched)	Live	• Limitation: 102 curated companies only. • Expansion: Add slugs; enable <code>includeCompensation=true</code> for salary data.
Workable	No-auth public API	No	apply.workable.com	4,137 stored (7,627 fetched)	Live	• Limitation: 89 curated companies; public widget API only, not full ATS. • Expansion: Add more verified company slugs.
Workday	No-auth public API	No	No fixed docs URL (keyless, per-tenant endpoint)	2,997 stored (2,998 fetched)	Live	• Limitation: Self-capped at 200 jobs/tenant; large tenants under-sampled. • Expansion: Raise per-tenant page cap; add more tenants.
Himalayas	No-auth public API	No	himalayas.app	2,253 stored (3,000 fetched)	Live	• Limitation: Self-capped at 3,000 of 90,000+; fixed page size (limit ignored). • Expansion: Raise page cap; use richer <code>/jobs/api/search</code> endpoint.
Arbetsförmedlingen	No-auth public API	No	jobsearch.api.jobteckdev.se	2,096 stored (2,100 fetched)	Live	• Limitation: Hard offset ≤2,000 ceiling (~2,100 of 40,000 reachable). • Expansion: None found; add in-loop error handling for resilience.
Adzuna	Free tier + paid metered	Yes, beyond free tier/quota	developer.adzuna.com	2,000 stored (2,000 fetched)	Live	• Limitation: Self-capped at 2,000/run; single-country (us) only. • Expansion: Raise page cap; query additional Adzuna country indexes.
The Muse	No-auth public tier	No	themuse.com	2,000 stored (2,000 fetched)	Live	• Limitation: Hard 100-page ceiling (HTTP 400) without an API key. • Expansion: Add an API key to raise the ceiling.
CareerJet	Free registration	No	careerjet.com	1,996 stored (2,000 fetched)	Live	• Limitation: Self-capped at 2,000/run (no ceiling found in testing). • Expansion: Raise page cap.
EURES	No-auth public feed	No	europa.eu/eures (undocumented)	1,681 stored (2,000 fetched)	Live	• Limitation: Hard ~10,000-result ceiling; self-capped at 2,000/run. • Expansion: Raise page cap toward the ~10,000 ceiling.
RemoteJobs.org	No-auth public API	No	remotejobs.org (docs link 404s)	1,500 stored (1,500 fetched)	Live	• Limitation: Real rate limiting observed (HTTP 429 ~page 23). • Expansion: None beyond current handling; partial results already preserved.
France Travail	Free self-service	No	francetravail.io	1,146 stored (1,150 fetched)	Live	• Limitation: Self-capped range ceiling (1,150), inherited from prior research. • Expansion: Re-confirm true ceiling; raise if larger window confirmed.
Jooble	Free registration	No	jooble.org	1,137 stored (1,137 fetched)	Live	• Limitation: Usable window ~1,100-1,200 jobs vs. much larger reported totals; no in-loop retry. • Expansion: Add in-loop error handling.
Get on Board	No-auth public tier	No	getonbrd.com	1,011 stored (1,261 fetched)	Live	• Limitation: None significant; paid Private API tier unused by design. • Expansion: None identified; already fetches to each filter's total_pages.
Taiwan Ministry of Labor	No-auth open data	No	taiwanjobs.gov.tw (data.gov.tw #44062)	1,000 stored (1,000 fetched)	Live	• Limitation: Hard 1,000-record cap; no working offset/page param found. • Expansion: None found; no further page accessible.
Findwork.dev	Free registration	No	findwork.dev	1,000 stored (1,000 fetched)	Live	• Limitation: Real rate limiting observed (HTTP 429 ~page 11). • Expansion: None beyond current handling; partial results already preserved.
Mustakbil.com	No-auth public feed	No	rss.mustakbil.com	500 stored (500 fetched)	Live	• Limitation: No pagination exists; fixed 500-item feed. • Expansion: None found.
Teamtaylor	No-auth public feed	No	docs.teamtaylor.com	494 stored (494 fetched)	Live	• Limitation: 25 curated companies; public RSS only, not credentialed Web API. • Expansion: Add more verified subdomains.
Recruitee (Tellent)	No-auth public API	No	docs.recruitee.com	394 stored (394 fetched)	Live	• Limitation: 21 curated companies; no pagination metadata in payload. • Expansion: Add more verified subdomains.
RemoteOK	No-auth public API	No	remoteok.com	100 stored (100 fetched)	Live	• Limitation: No pagination exists; fixed ~100-job snapshot. • Expansion: None found.
MyJobMag	No-auth public feed	No	myjobmag.com	100 stored (100 fetched)	Live	• Limitation: No pagination (robots.txt blocks query strings); fixed 100 items. • Expansion: Add other MyJobMag country feeds (Ghana/Kenya/SA/UK).

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Jobs Retrieved (Current Impl.)	Status	Implementation Notes
Jobicy	No-auth public API	No	jobicy.com	100 stored (100 fetched)	Live	• Limitation: No pagination; hard 100-job cap on <code>count</code>. • Expansion: None found.
We Work Remotely	No-auth public feed	No	weworkremotely.com	99 stored (100 fetched)	Live	• Limitation: No pagination; fixed ~100-item feed. • Expansion: Add per-category feeds for broader coverage.
4dayweek.io	No-auth public feed	No	4dayweek.io	50 stored (50 fetched)	Live	• Limitation: No pagination; richer <code>/api/jobs</code> endpoint blocked by robots.txt. • Expansion: None; richer endpoint deliberately excluded.
SerpApi (Google Jobs)	Free tier + paid metered	Yes, beyond free tier/quota	serpapi.com	50 stored (50 fetched)	Live	• Limitation: Metered API; self-capped at 50 jobs/run to conserve quota. • Expansion: Raise page cap if a larger quota is approved.
OpenWebNinja (JSearch)	Free tier + paid metered	Yes, beyond free tier/quota	openwebninja.com	49 stored (49 fetched)	Live	• Limitation: Metered API; self-capped at 5 pages to conserve quota. • Expansion: Raise page cap if a larger quota is approved.
Working Nomads	No-auth public API	No	workingnomads.com	33 stored (33 fetched)	Live	• Limitation: No pagination; fixed ~36-job snapshot; no RSS alternative found. • Expansion: None found.
Remotive	No-auth public API	No	remotive.com	30 stored (30 fetched)	Live	• Limitation: No pagination; filter params confirmed non-functional. • Expansion: None found.
Freshersworld	No-auth public feed	No	freshersworld.com	30 stored (30 fetched)	Live	• Limitation: No pagination; rolling ~30-item window. • Expansion: None found.
Jobspresso	No-auth public feed	No	jobspresso.co	20 stored (20 fetched)	Live	• Limitation: Deeper pagination exists but blocked by robots.txt. • Expansion: None; deeper pagination deliberately unused.
MyCareersFuture	No-auth public API	No	mycareersfuture.gov.sg (undocumented)	20 stored (2,000 fetched)	Needs review	• Limitation: <code>limit</code> param ignored (fixed 20/page); self-capped at 2,000/run. • Expansion: Raise page cap; no hard API ceiling found.
NoDesk	No-auth public feed	No	nodesk.co	10 stored (10 fetched)	Live	• Limitation: No pagination; fixed 10-item feed. • Expansion: None found.
Hasjob	No-auth public feed	No	hasjob.co	6 stored (6 fetched)	Live	• Limitation: No pagination; inherently small volume (community board). • Expansion: None found.
HK Gov Vacancies	No-auth open data	No	csb.gov.hk (data.gov.hk)	1 stored (58 fetched)	Needs review	• Limitation: Static dataset; no per-posting URL (collapses on dedup). • Expansion: Use a non-URL dedup key to reduce duplicate collapse.
TheirStack	Per-credit metered (1 credit/job)	Yes	theirstack.com	0 stored (0 fetched)	Blocked — billing	• Limitation: Bills per job returned; blocked by account billing (HTTP 402). • Expansion: Resolve billing; then raise job cap and add retry helper.
Trade Me Jobs	Free (registered app)	No	developer.trademe.co.nz	0 stored (0 fetched)	Built, uncredentialed	• Limitation: Built from API spec only, unexercised live; missing credentials (0 jobs). • Expansion: Provision consumer key/secret; complete production approval.
CareerOneStop	Free registration	No	careeronestop.org	0 stored (0 fetched)	Built, uncredentialed	• Limitation: Hard ~750-result ceiling; missing credentials (0 jobs). • Expansion: Provision user ID/API token (free registration).
Fantastic.jobs	Free trial → paid metered (~\$1/1,000 jobs)	Yes, after trial ends	developer.fantastic.jobs	0 stored (0 fetched)	Credentialed, 0 result	• Limitation: Credentialed but 0 jobs; cause not confirmed. • Expansion: Verify account/trial status; add <code>/v1/active-jobs</code> sibling endpoint.
Arbeitsnow	No-auth public API	No	arbeitsnow.com	0 stored (0 fetched)	0 in latest run	• Limitation: 0 jobs in latest run (cause unconfirmed); no retry/error handling. • Expansion: Add retry/error handling; investigate 0-job run.

4.2 NOT IMPLEMENTED PROVIDERS (222)

Grouped by research classification at the time of this project. **Registration/Approval Required** and **Needs Live Verification** providers have a real, plausible access path (Section 6, Recommendations); **Not Viable** providers were rejected on structural grounds (no public API, enterprise-only, wrong data-flow direction, deprecated, or a dataset rather than a live API).

Registration/Approval Required (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
CBOP - Centralna Baza Ofert Pracy (Poland public employment offices)	Free	No	praca.gov.pl	Potentially large - national public employment office coverage	Poland's national public-employment-office job database. Real SOAP web service returning batches of up to 1,000 live offers per file, but capped at 20 calls/cycle and only reachable during a 17:00-07:00 window -...
Saramin Open API (사람인 오픈 API)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	saramin.co.kr	Unknown - Saramin is one of Korea's largest commercial job boards but no public...	One of Korea's largest commercial job boards. Real documented API, but gated behind an application/approval step, explicitly labeled beta with a 500-calls/day cap - usable, just slower to unlock and rate-constrained.
VDAB Vacature API (Belgium - Flanders public employment service)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large - Flanders regional coverage	Flanders' public employment service vacancy API - free but requires an application form and an exploratory onboarding meeting with VDAB before credentials are issued. Real read API, just a slower registration path than...
Work24 (formerly Worknet/워크넷) Job Listings Open API - Korea Employment Information Service	Free	No	Not publicly available	Potentially large - South Korea's national public employment portal (renamed 고용24 in...	South Korea's national public job bank, verified live on data.go.kr with real-time XML/JSON updates and a free key. Caveat: license text restricts use to non-commercial, attributed, unmodified use - confirm terms with...

Needs Live Verification (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
GOV.UK Find a Job (DWP)	Free	No	Not publicly available	Potentially large	UK's national government job service; no confirmed public read/search API for third parties was found - only an employer-side SFTP bulk-upload path for posting vacancies. Worth one more direct check with DWP given its...
Job-Room Jobs API (SECO, Switzerland)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	Switzerland's biggest job platform, but the API is built primarily for employers/ATS submitting ads under a reporting-obligation rule - whether third-party aggregators can get read-only access is unconfirmed in the...
JobsPikr	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Unknown (volume not published)	Commercial aggregator (incl. Naukri/Monster-sourced India feeds) with a real API and docs, but pricing is custom/quote-based rather than a published self-serve rate - confirm actual signup process and cost before...
Portal del Empleo / Servicio Nacional de Empleo (STPS Mexico)	Free	No	Not publicly available	Potentially large	Mexico's national employment service publishes datasets on the CKAN-based datos.gob.mx open-data platform, but a live, current-vacancy-specific API endpoint for the job board itself was not directly confirmed - may...

Not Viable at Research Time (214)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
104 Job Bank (104人力銀行)	Not applicable, no public API	Not applicable, no public API	104.com.tw	~1.1 million (110萬, Feb 2026)	No publ site/port jobseek login; nc access i anywhe
1111 Job Bank (1111人力銀行)	Not applicable, no public API	Not applicable, no public API	1111.com.tw	Not publicly available	No publ site/port jobseek login; nc access i anywhe
51job (前程无忧 / Qiancheng Wuyou)	Not applicable, no public API	Not applicable, no public API	51job.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
AMS eJob- Room (Austria public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
APEC (Association Pour l'Emploi des Cadres, France)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
APS Jobs (Australian Public Service careers)	Not applicable, no public API	Not applicable, no public API	apsjobs.gov.au	Moderate	No publ site/port jobseek login; nc access i anywhe
Adecco India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Agencia Publica de Empleo SENA (APE)	Not applicable, no public API	Not applicable, no public API	edu.co	Potentially large	No publ site/port jobseek login; nc access i anywhe
AllJobs.co.il	Not applicable to data access (fee is for employers posting jobs on	Not applicable to data access (see Pricing Model column)	AllJobs.co.il	Potentially large	No publ site/port jobseek login; nc

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
	this platform, not...				access i anywhe
Apideck	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	apideck.com/pricing	Not publicly available	Paid mix re-wrap: (several Greenh Workda integrat free). Re enterpri
Apify	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	apify.com/pricing	Not publicly available	Generic infrastru not a job using it and mai custom ourselfe project i more pr
Apna	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No publ party ag
Appcast	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	appcast.io/how-programmatic-recruitment-advertising-improves-cost-per-hire/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Aura (by Aura Intelligence)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Proprietary workforce database spanning 20M+ companies (billions of data points;...	Enterpri market-i SaaS sc contract serve jo feed.
BMET / Ami Probashi (Bangladesh Overseas Employment)	Not applicable, no public API	Not applicable, no public API	amiprobashi.com	Potentially large	No publ site/port jobseek login; nc access i anywhe
BOSS Zhipin (Kanzhun Limited, NASDAQ: BZ)	Not applicable, no public API	Not applicable, no public API	zhipin.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
BambooHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterpri pricing/ε behind ε partner- process develop

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Bayt.com	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Bayt.com	Potentially large	No publ site/port jobseek login; nc access i anywhe
Bdjobs.com	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
Bolsa Nacional de Empleo (BNE / SENCE, Chile)	Free	No	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
Bossjob	Free	No	bossjob.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Bright Data	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	brightdata.com/products/web-scraper/jobs-scraper	116.8M+ records (general Jobs dataset); LinkedIn Jobs dataset specifically 62.4M+	Generic infrastru not a jot using it and mai custom ourselve project i more pr
BrighterMonday	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	brightermonday.co.ke	Potentially large	No publ site/port jobseek login; nc access i anywhe
Broadbean (Veritone Hire)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	broadbean.com/product-suite/job-distribution-platform/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Built In	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech/startup; not quantified)	No publ site/port jobseek login; nc access i anywhe

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Bumeran	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
CEIPAL	Not publicly available	Not publicly available	ceipal.com/pricing	Not publicly available	No publ site/port jobseek login; nc access i anywhe
CV-Library	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
CWJobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	CWJobs.co.uk	12,000+ jobs (per cwjobs.co.uk About Us page)	No publ site/port jobseek login; nc access i anywhe
CareerBuilder	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerbuilder.com/	Not publicly available	No publ site/port jobseek login; nc access i anywhe
CareerBuilder / Monster (Talent Network distribution)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	careerbuilder.com/talent-network	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
CareerLink Vietnam	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
CareerOne (Australia)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerone.com.au	100,000+ (live jobs)	Feed-in; distribut only - nc endpoin data as
Careers24	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	careers24.com	Potentially large	No publ site/port jobseek login; nc

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					access i anywhe
Careers@Gov	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (SG public sector; not quantified)	No publ site/port jobseek login; nc access i anywhe
Catho Developer API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
China MOHRSS National Public Employment Service Platform (全国公共招聘服 务平台)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Civil Service Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (not quantified)	No publ site/port jobseek login; nc access i anywhe
Claro Analytics (WilsonHCG)	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	claroanalytics.com/	Real-time job-postings and 500M+ global talent profiles	Enterpri market-i SaaS sc contract serve jo feed.
Cliclavoro / Ministero del Lavoro Open Data (Italy)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Computrabajo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Coresignal	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	Not publicly available	399M+ multi-source records (incl. India)	Enterpri vendor s contact, publishe price - s the othe data ver
Cutshort	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (tech; board size not public)	No publ site/port jobseek

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					login; no access i anywhere
Darwinbox	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterpri pricing/ε behind ε partner-process develop
Datapeople (by Payscale)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	datapeople.io/	Not publicly available (N/A (analyzes customer job postings, not a broad aggregation...	Enterpri market-i SaaS sc contract serve jo feed.
Dice	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech niche; not quantified)	No publ site/port jobseek login; no access i anywhere
Diffbot	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	diffbot.com/pricing	Not publicly available	Generic infrastru not a job using it and mai custom ourselve project i more pr
Draup	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	draup.com/	Global labor-market and job-postings data (aggregated multi-source)	Enterpri market-i SaaS sc contract serve jo feed.
Drushim	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	Drushim.co.il	Potentially large	No publ site/port jobseek login; no access i anywhere
Egypt Ministry of Labour (formerly Ministry of Manpower) Job Portal	Not applicable, no public API	Not applicable, no public API	labour.gov.eg	Not publicly available	No publ site/port jobseek login; no access i anywhere
Elemplo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Very large (CO; not quantified)	No publ site/port jobseek login; no access i anywhere

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Eluta	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Employment Services of South Africa (ESSA) - Dept. of Employment and Labour	Free	No	labour.gov.za	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Forasna	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	forasna.com/employer/pricing	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Foundit (Monster India)	Not applicable, no public API	Not applicable, no public API	Not publicly available	~800k+ (board size; no public access)	No publ party ag
Freshteam	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterpri pricing/ε behind ε partner-process develop
Glassdoor Jobs API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	glassdoor.com/developer/index.htm	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Glints	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Google Cloud Talent Solution (Job Search API)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	google.com/talent-solution/pricing	Not publicly available (N/A (you populate it with your own jobs; not an aggregation...	This is ε matchin engine y with you postings source ε third-par Enterpri infrastru not a job
Google for Jobs	Not applicable to data access (fee is for employers posting jobs on	Not applicable to data access (see Pricing Model column)	schema.org	Unknown (indexes open web)	Advertis distribut network employe postings aggrega

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
	this platform, not...				postings read AP
Greenwich.HR (WageScape)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	amazon.com/marketplace/pp/prodview-t5fvgfurgypxg	~5M new US jobs/month; ~70% of all open US jobs; 1.5M+ hiring organizations	Enterpri market-i SaaS sc contract serve jo feed.
GreythHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No publ site/port jobseek login; nc access i anywhe
GulfTalent	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	gulftalent.com	Potentially large	No publ site/port jobseek login; nc access i anywhe
Gupy Public API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	gupy.io	~90,000 job vacancies published monthly on the Gupy R&S platform (official figure)	No publ site/port jobseek login; nc access i anywhe
HackerEarth Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (board size not public)	No publ site/port jobseek login; nc access i anywhe
HasData	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	hasdata.com/prices	Not publicly available	Generic infrastru not a job using it and mai custom ourselve project i more pr
Hello Work Job Information API (ハローワーク求人情報提供サービス, MHLW)	Free	No	Not publicly available	Potentially large - nationwide public employment service covering hundreds of thousands...	Japan's docume job API - registrat to licens placeme local go training generic does no
HelloWork ATS Partner API (France)	Free, but partner/approval access required	No, once approved, but partner/approval	Not publicly available	Not publicly available	No publ site/port jobseek

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
	(no self-serve signup)	access is itself the...			login; no access i anywhere
Hired.com (LHH Recruitment Solutions)	Not applicable, no public API	Not applicable, no public API	Hired.com	0	No publ site/port jobseek login; no access i anywhere
Hirist	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No publ site/port jobseek login; no access i anywhere
Hong Kong Labour Department Interactive Employment Service (IES, jobs.gov.hk)	Not applicable, no public API	Not applicable, no public API	www.jobs.gov.hk	Not publicly available	No publ site/port jobseek login; no access i anywhere
Horsefly Analytics	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	horseflyanalytics.com/	1 trillion+ data points; millions of job postings; supply/demand across 60+ countries	Enterpri market-i SaaS sc contract serve jo feed.
IEFPOne / netEmprego (Portugal public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	5,913 ofertas de emprego (live count)	No publ site/port jobseek login; no access i anywhere
Indeed Hiring Lab	Free	No	indeed.com	Not publicly available (N/A (publishes indices/aggregates, e.g. Job Postings Index, Wage...	Indeed's research publishe labor-ma indices/l (mirrore not indiv records. Researc product, listings f
Indeed India	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Huge (not accessible - partner-only)	Enterpri access; a generi
Indeed PLUS Job Distribution API (Recruit Holdings, Japan)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large - aggregates postings across Japan's major job boards distributed via...	Advertis distribut network employe postings aggrega

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					postings read AP
InfoJobs (Spain)	Not applicable, no public API	Not applicable, no public API	Not publicly available	68,383 ofertas (live count, Spain, as of July 2026)	No publ site/port jobseek login; nc access i anywhe
Instahyre	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No publ site/port jobseek login; nc access i anywhe
Internshala	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large (intern/fresher; board size not public)	No publ site/port jobseek login; nc access i anywhe
Israeli Employment Service (Sherut HaTaasuka / taasuka.gov.il)	Free	No	taasuka.gov.il	Potentially large	No publ site/port jobseek login; nc access i anywhe
JSearch (RapidAPI, by OpenWeb Ninja)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	rapidapi.com/letsrape-6bRBa3QguO5/api/jsearch	Not publicly available	Generic aggrega a dedicē paid, an sources integratē
Jadarat (Saudi National Employment Portal / HRDF)	Free	No	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
Job Bank (ESDC) feed / Open Gov dataset	Free	No	Not publicly available	Very large (100,000+)	Canada bank: th requires approva fallback a month missing employe names & apply UI fields th schema
Job Market Finland / Tyomarkkinatori (TE-palvelut, Finland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					access i anywhe
JobKorea	Not applicable, no public API	Not applicable, no public API	Not publicly available	~209,637 (live listings)	No publ site/port jobseek login; nc access i anywhe
JobStreet / JobsDB (SEEK group)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Very large if accessible (not public)	SEEK-g boards f enterpri access, path (ca marks 'I Recomr
JobThai	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobthai.com	~121,000 active jobs (121,337 shown live on jobthai.com)	No publ site/port jobseek login; nc access i anywhe
Jobberman	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	jobberman.com	Potentially large	No publ site/port jobseek login; nc access i anywhe
Jobcase (MaxRecruit)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobcase.com/products/maxrecruit	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Jobcase Platform API	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jobcase.com/	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Jobg8	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jobg8.com/Faq.aspx?contentid=10874	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Jobillico	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No publ site/port jobseek login; nc

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					access i anywhe
Jobindex (Denmark)	Not applicable, no public API	Not applicable, no public API	Not publicly available	33,600 jobs (live) / 30,000+ job ads	No publ site/port jobseek login; nc access i anywhe
JobisJob (LIFULL Connect)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobboardfinder.com/jobboard-jobisjob-usa	~9,000,000 (monthly, cross-country)	Advertis distribut network employe postings aggrega postings read AP
Jobnet.dk (Denmark public employment portal)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Jobrapido	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	jobrapido.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Jobs.cz	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterpri pricing/ε behind ε partner-process develop
JobsDB Hong Kong (SEEK Group) / CTgoodjobs	Not applicable, no public API	Not applicable, no public API	hk.jobs	~32,700 (JobsDB HK) / ~32,000 (CTgoodjobs)	No publ site/port jobseek login; nc access i anywhe
JobsEQ (Chmura Economics & Analytics)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	chmura.com/jobseq	Real-time job-posting dataset with international coverage	Enterpri market-i SaaS sc contract serve jo feed.
JobsIreland.ie (Ireland public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	4,806 vacancies (live count)	No publ site/port jobseek login; nc access i anywhe
Jobsite	Enterprise	Yes, enterprise/sales	Jobsite.co.uk	280,000+ live job adverts (per	No publ site/port

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
		pricing, no self-serve free tier		jobsite.co.uk About Us page)	jobseek login; nc access i anywhe
JoinVision (JobCloud HR Tech)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	crunchbase.com/organization/joinvision	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Jora (Australia)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jora.com	Potentially large	No publ site/port jobseek login; nc access i anywhe
Joveo	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	joveo.com/programmatic-job-advertising-platform/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
JustRemote	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	justremote.co/remote-jobs	2,000+ hidden remote jobs	No publ site/port jobseek login; nc access i anywhe
Kalibr	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Karir.com	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Keka	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No publ site/port jobseek login; nc access i anywhe
Kelly Services India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					access i anywhe
Kemnaker Karirhub (SIAPkerja)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
Kombo.dev	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	kombo.dev/pricing	Not publicly available	Paid mic re-wrap: (several Greenh Workda integrat free). R enterpri:
Ladders	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (niche; not quantified)	No publ site/port jobseek login; nc access i anywhe
Lensa	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	lensa.com/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Lightcast (formerly Emsi Burning Glass) Job Postings API	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	lightcast.dev/apis/job-postings	Billions of postings aggregated from 160,000+ sources (18B+ labor-market data points)	Enterpri market-i SaaS sc contract serve jo feed.
Lightcast Jobs Canada	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	Paid lab intellige sold thro process serve de signup; marks 'I Recomr
LinkUp	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	linkup.com	Millions of postings indexed daily from 80,000+ employer websites	Enterpri market-i SaaS sc contract serve jo feed.
LinkedIn Jobs	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	Not publicly available	Huge (not accessible - partner-only)	Enterpri program serve de for job search/

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
MOHRE UAE Federal Careers / iRecruitment Portal	Not applicable, no public API	Not applicable, no public API	mohre.gov.ae	Not publicly available	No publ site/port jobseek login; nc access i anywhe
MP Rojgar	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No publ site/port jobseek login; nc access i anywhe
MahaJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No publ site/port jobseek login; nc access i anywhe
ManpowerGroup India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Mantiks	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	mantiks.io/pricing	Not publicly available	Generic aggrega a dedica paid, an sources integrat
Merge.dev	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	merge.dev/pricing/unified	Not publicly available	Paid mix re-wrap: (several Greenh Workda; integrate free). Re enterpri:
Mitula (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	lifullconnect.com/brands/mitula/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Monster	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	monster.com/solutions/talent-management/partner-with-monster/	Not publicly available	No publ site/port jobseek login; nc access i anywhe

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
NAVTTTC National Employment Exchange Tool (NEXT / jobs.gov.pk)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
NEAIMS - National Employment Authority Kenya	Free	No	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
National Career Service (NCS)	Free	No	data.gov.in	Large in aggregate	India's N periodic governn (data.gc per-job : stale sn real-time Doesn't architect
National Directorate of Employment (NDE) Nigeria	Not applicable, no public API	Not applicable, no public API	nde.gov.ng	Not publicly available	No publ site/port jobseek login; nc access i anywhe
National Job Portal (NJP) - Pakistan	Not applicable, no public API	Not applicable, no public API	njp.gov.pk	Small-to-moderate	No publ site/port jobseek login; nc access i anywhe
National Job Portal - Bangladesh (jobs.gov.bd)	Not applicable, no public API	Not applicable, no public API	jobs.gov.bd	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Naukri	Not applicable, no public API	Not applicable, no public API	Not publicly available	~1M+ (board size; no public access)	No publ access i undocur area an restricte
Naukrigulf	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	naukrigulf.com	Potentially large	No publ site/port jobseek login; nc access i anywhe

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Nexxt (formerly Beyond.com)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Beyond.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Nimble (Nimbleway)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	nimbleway.com/pricing	Not publicly available	Generic infrastru not a jot using it and mai custom ourselve project i more pr
OCC Mundial	Not applicable, no public API	Not applicable, no public API	Not publicly available	137,925 active vacancies ("vacantes activas en occ.com.mx", as of Oct 18, 2024);...	No publ site/port jobseek login; nc access i anywhe
Outsourcely	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	outsourcely.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Oxylabs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	oxylabs.io/products/scraper-api/web/pricing	Not publicly available	Generic infrastru not a jot using it and mai custom ourselve project i more pr
PandoLogic (pandoiQ) / Veritone Hire	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	pandologic.com/solutions/programmatic-advertising-pandoiq/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Pangian	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	pangian.com/feed/?post_type=job_listing	Varies (curated remote listings)	pangian (301) to You for l Pangian notice o Pages (pangiar confirme other pa /about/, candida

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
People Data Labs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	peopledatalabs.com/docs/job-posting-data-overview	Not publicly available	Generic aggrega a dedicæ paid, an sources integrat
PhilJobNet (DOLE Bureau of Local Employment)	Free	No	philjobnet.gov.ph	Potentially large	No publ site/port jobseek login; nc access i anywhe
Piloterr	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	piloterr.com/pricing	Not publicly available	Generic infrastru not a job using it and mai custom ourselve project i more pr
Pnet	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	pnet.co.za	Potentially large	No publ site/port jobseek login; nc access i anywhe
Portal Empleo Argentina (Ministerio de Capital Humano / ex-Trabajo)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
Pracuj.pl (Poland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	762,827 job offers published in 2025	No publ site/port jobseek login; nc access i anywhe
PredictLeads	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	predictleads.com/pricing	270M+ records since 2018; ~9.8M active jobs	Generic aggrega a dedicæ paid, an sources integrat
Profesia	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterpri pricing/ε behind ε partner-process develop
Public Service Commission of	Not applicable, no public API	Not applicable, no public API	psc.gov.lk	Not publicly available	No publ site/port jobseek

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Sri Lanka (psc.gov.lk)					login; no access i anywhere
Quess Corp	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; no access i anywhere
Radancy	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	radancy.com/en/programmatic-adtech/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Randstad India	Not applicable, no public API	Not applicable, no public API	Not publicly available	1,394 jobs (live count on official site)	No publ site/port jobseek login; no access i anywhere
Recruit CRM	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterpri pricing/ε behind ε partner-process develop
Recruitics	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	recruitics.com/programmatic-job-advertising	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Remote First Jobs (Dynamite Jobs)	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	remotefirstjobs.com/rss	Varies (curated remote-first listings)	remotefi behind ε Cloudfla challeng path tes robots.b confirme 403 with moment challeng a...
Remote.co	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	remote.co/feed	Not publicly available	No publ site/port jobseek login; no access i anywhere

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Revelio Labs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	reveliolabs.com	1.1B+ workforce profiles plus job postings, transitions, sentiment, layoffs	Enterpri market-i SaaS sc contract serve jo feed.
Rikunabi / Mynavi (Recruit Co. / Mynavi Corporation)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Rojgar Sangam (UP)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No publ site/port jobseek login; nc access i anywhe
Rozee.pk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
SEEK (SEEK.com.au / SEEK.co.nz)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	SEEK.com.au	Potentially large	Feed-in; distribut only - nc endpoin data as
SEPE Open Data Portal (Spain public employment service)	Free	No	Not publicly available	Not publicly available	Spain's publishe statistic; open-da confirm API for c vacanci known t scraper depreca
SINE Aberto (Sistema Nacional de Emprego / Ministerio do Trabalho e Emprego)	Free	No	dados.gov.br/dataset/sine-aberto	Potentially large	Brazil su data-sh; October (CODEF 956/202 LGPD p complia no activ integrate now.
ScraperAPI	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	scraperapi.com/pricing	Not publicly available	Generic infrastru not a job using it and mai custom

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					ourselfe project i more pr
Scrapingdog	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	scrapingdog.com/pricing	Not publicly available	Generic infrastru not a job using it and mai custom ourselfe project i more pr
Shine	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No publ site/port jobseek login; nc access i anywhe
SmartDreamers	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	smartdreamers.com/pricing	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Snagajob	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	snagajob.com/	Not publicly available	No publ site/port jobseek login; nc access i anywhe
StepStone	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	stepstone.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Superset	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (campus; per-institution, not aggregatable)	Enterpri pricing/ε behind ε partner- process develop
Talent.com	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Talent.com	40 million+ jobs (global figure stated on talent.com homepage)	No publ site/port jobseek login; nc access i anywhe
TalentNeuron (Gartner)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	talentneuron.com/solutions/platform	Real-time labor data covering ~90% of world GDP; 40TB	Enterpri market-i SaaS sc contract

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
				normalized data, 3B+ profiles,...	serve jo feed.
Talroo	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	talroo.com/pro/	Millions (per Talroo publisher marketing)	Advertis distribut network employe postings aggrega postings read AP
TeamLease	Not applicable, no public API	Not applicable, no public API	Not publicly available	23,082+ active job vacancies (live count on official site)	No publ site/port jobseek login; nc access i anywhe
Techmap.io	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	jobdatafeeds.com	292M+ available jobs	Generic aggrega a dedica paid, an sources integrat
Telangana T-Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No publ site/port jobseek login; nc access i anywhe
Textkernel Jobfeed / Market IQ	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	textkernel.com/products-solutions/labour-market-insights/market-iq/	Large multi-country vacancy corpus (millions of postings, deduplicated/enriched)	Enterpri market-i SaaS sc contract serve jo feed.
Thailand Department of Employment - Smart Job Center (ไทยมีงานทำ)	Free	No	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe
The Burning Glass Institute	Free	No	burningglassinstitute.org/	Not publicly available (N/A (research using Lightcast and other data))	Enterpri market-i SaaS sc contract serve jo feed.
TimesJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No publ site/port jobseek login; nc access i anywhe
TopCV	Paid (metered/flat;	Yes, paid from the outset, no	Not publicly available	Potentially large	No publ site/port

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
	see Cost Details column source for exact terms)	free tier documented			jobseek login; nc access i anywhe
TotalJobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Totaljobs.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Triplebyte	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	triplebyte.com/	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Trovit (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	trovit.com/feed-in-jobs/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
UWV Open Match Data (Netherlands)	Free	No	Not publicly available	Not publicly available (N/A - aggregated statistics only, not individual postings)	UWV (D employr publishe labor-m: datasets live job p per-job i in the d:
Unified.to	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Not publicly available	Not publicly available	Paid mix re-wrap: (several Greenh: Workda: integrate free). R: enterpri:
Unstop	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (campus/intern; not a job-listing API)	No publ site/port jobseek login; nc access i anywhe
Upward.net (a Jobcase company)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	upward.net/employers	Not publicly available	Advertis distribut network employe postings aggrega postings read AP

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
Vagas for Business API (Vagas.com.br)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Vagas.com	Not publicly available	No publ site/port jobseek login; nc access i anywhe
VietnamWorks Open API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	vietnamworks.com	Potentially large	Partner Require approve partners develop
WUZZUF	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	wuzzuf.net	Potentially large	No publ site/port jobseek login; nc access i anywhe
Welcome to the Jungle	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Wellfound	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (startup roles; board size not public)	No publ site/port jobseek login; nc access i anywhe
WhatJobs	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	whatjobs.com/job-aggregators	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
Work and Income / Find a Job (MSD, New Zealand)	Not applicable, no public API	Not applicable, no public API	workandincome.gov	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Work at a Startup (Y Combinator)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	workatastartup.com	Unknown (thousands across 1,000+ YC companies)	No publ site/port jobseek login; nc access i anywhe
WorkIndia	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No publ site/port jobseek login; nc

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					access i anywhe
Workato	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	workato.com/pricing	Not publicly available	Paid mix re-wrap: (several Greenh Workda integrat free). Re enterpri:
Workforce Australia	Free	No	workforceaustralia.gov.au	Potentially large	The pub write/m (employ manage OAuth2 Busines public j read/se exists to from.
Workopolis	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
WowJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Xing Jobs	Enterprise	Yes, enterprise/sales pricing, no self- serve free tier	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
XpressJobs (xpress.jobs)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No publ site/port jobseek login; nc access i anywhe
ZipAlerts (ZipRecruiter TrafficBoost)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	zipalerts.com/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
ZipRecruiter API	Enterprise	Yes, enterprise/sales	Not publicly available	Large (not quantified)	Enterpri pricing/ε behind ε

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
		pricing, no self-serve free tier			partner-process develop
Zoho Recruit	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterpri pricing/ε behind ε partner-process develop
ZonaJobs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	zonajobs.com.ar	Not publicly available	No publ site/port jobseek login; nc access i anywhe
Zyte (formerly Scrapinghub)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	zyte.com/zyte-api/pricing.html	Not publicly available	Generic infrastru not a job using it and mai custom ourselve project i more pr
data.gov.in OGD platform	Free	No	Not publicly available	Varies (per-resource)	General open-gc portal sp sectors; specific, resource per-resc rather th postings
eQuest	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	equest.com/	Not publicly available	Advertis distribut network employe postings aggrega postings read AP
iimjobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (management roles; board size not public)	No publ site/port jobseek login; nc access i anywhe
jobs.ch (Switzerland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	45,825 jobs (live)	No publ site/port jobseek login; nc access i anywhe
karriere.at (Austria)	Not applicable, no public API	Not applicable, no public API	github.com/karriereat	13,400+ jobs (live)	No publ site/port

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Implem
					jobseek login; nc access i anywhe
topjobs.lk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No publ site/port jobseek login; nc access i anywhe

5. Internship Requirement Coverage

2. FINAL PROJECT REPORT

This document, together with the companion technical report `Final_Project_Report_v2.md`, is the final project report.

3. APIS/SERVICES USED

46 providers implemented (Section 3); 268 researched in total (Section 4).

4. COMPLETED VS. PENDING

See the [Project Status table](#) (Section 2).

5. FULLY / PARTIALLY / NOT STARTED

See the [Project Status table](#) (Section 2).

6. PRICING MODEL (FREE / FREE TRIAL / PAY-AS-YOU-GO / SUBSCRIPTION / ENTERPRISE)

Pricing Model (Implemented Providers)	Count	Providers
Free (no cost, ever)	41	40 fully free/no-auth or free-registration providers, plus Trade Me Jobs (free, but approval-gated)
Free Trial (time-limited, then paid)	1	Fantastic.jobs (7-day trial, then metered)
Pay-as-you-go / Metered (has a paid, usage-based component)	5	Adzuna, SerpApi, OpenWebNinja, TheirStack, Fantastic.jobs
Subscription (flat recurring fee)	0	Not used by any implemented provider
Enterprise (sales-negotiated contract)	0	Not used by any implemented provider — 15 enterprise-only candidates were identified in research and not implemented (Section 4.2)

Full per-provider pricing model is in Section 4.1. The 222 not-implemented providers' pricing (where verifiable) is in Section 4.2.

7. PAYMENT REQUIRED AFTER FREE TIER

Answer	Count (of 46 implemented)
No (free with no paid tier)	41
Yes, beyond a free tier/quota, or after a trial	5

8. PRICING / DOCUMENTATION LINKS

All 46 implemented providers have an official reference URL, quoted directly from source code (Section 4.1). Of the 222 not-implemented providers, a documented reference (domain or URL) was found in this project's research for **166 of 268 providers overall**; for the remainder, the reference cell reads "Not publicly available" rather than a guessed link (Section 4.2).

9. NUMBER OF JOBS

Metric	Value
Total unique jobs stored (PostgreSQL)	184,506
Total jobs fetched, all providers, pre-dedup	200,727
Distinct companies represented	18,367

Top providers by volume are in Section 3.3. Approximate job counts for not-implemented providers (where publicly known) are in Section 4.2.

10. REGION COVERAGE

Region	Dedicated Providers	Notes
United States	4	USAJOBS, CareerOneStop, Findwork.dev, Adzuna
Canada	0	No dedicated source; incidental coverage only via global ATS boards
United Kingdom	2	Reed, NHS Jobs
Europe (non-UK)	5	Bundesagentur für Arbeit, Arbetsförmedlingen, France Travail, EURES, NAV Arbeidsplassen
India	2	Freshersworld, Hasjob
LATAM	1	Get on Board
APAC	4	MyCareersFuture, Taiwan MOL, HK Gov Vacancies, Trade Me
Africa	1	MyJobMag
Middle East	0	No single-country dedicated source (Mustakbil spans 20+ countries, counted as global)
Global / Remote / Multi-country	27	All 6 ATS platforms plus 21 aggregator/remote-board providers

Resulting job distribution is US-heavy (47.3%), then UK (14.3%) and Germany (6.9%) — a sourcing-mix effect from which countries have dense curated ATS company lists, not a normalization defect. Canada remains the clearest unresolved regional gap.

11. JOB CATEGORIES

No standardized, cross-provider job category exists in this dataset, and this cannot be supported without inventing one.

Verified directly against all 46 providers' `normalize()` implementations:

- **0 of 46** providers expose a job-category field that means the same thing across providers.
- **23 of 46** expose a provider-specific department, team, business-unit, occupation, or function field (e.g. SmartRecruiters' `function`, Taiwan Ministry of Labor's occupation code, USAJOBS's federal job series) — each proprietary to that one provider.
- **23 of 46** expose no category-like field at all (confirmed in code, e.g. Reed's module comment: "no job-category/employment-type field at all").

Evidence that these fields cannot be merged: the stored tag “Engineering” comes from SmartRecruiters’ `function` field (6,797 jobs), Greenhouse’s `departments` field (1,154 jobs), and Ashby’s `department / team` field (1,136 jobs) — three structurally unrelated fields that happen to use the same English word. Treating them as one taxonomy would silently merge unrelated data. This project’s schema stores all of this in a single free-text `tags` array; a real crosswalk would require manually mapping each provider’s raw values, which has not been done.

12. DATA QUALITY

Dimension	Finding
Missing fields	Confirmed per-provider in code — e.g. Reed exposes no employment-type field at all; no provider exposes usable salary data
Company info	43 of 46 providers give a direct company-name field; 3 derive it (Ashby, Lever, Workday)
Duplicate rate	8.08% overall (see item 14 below)
Stale postings	11.9% of stored jobs are over 1 year old

13. JOB FRESHNESS

Metric	Value
Jobs with a posted date recorded	184,129 of 184,506 (99.8%)
Median posting age	22 days
Mean posting age	229 days (skewed by a long tail of old postings on some boards)
Posted within 30 days of snapshot	108,003 (58.6%)
Older than 1 year	21,915 (11.9%)

Refresh frequency and expired-job detection are **not verified during this project** — the dataset reflects one ingestion run, not a recurring schedule, so there is no interval or diff to measure yet.

14. DUPLICATE ANALYSIS

Metric	Value
Total in-run duplicates skipped (all providers)	16,221
Total fetched, pre-dedup	200,727
Overall deduplication rate	8.08%
Cross-provider duplicate URLs remaining	0

Highest duplicate rates: MyCareersFuture (99.0%, `limit` param silently ignored by the provider), HK Gov Vacancies (98.3%, every job shares one generic portal URL), NAV Arbeitsplassen (56.8%, a change-feed rather than a current-postings search), Workable (45.8%, the same job posted to multiple cities, correctly collapsed).

15. RATE LIMITS

Limit type	Examples
Hard result-window ceiling	USAJOBS and Bundesagentur für Arbeit (10,000 records); Reed (~9,900, returns HTTP 500 beyond it)
Monthly quota	SerpApi and OpenWebNinja (200 requests/month free tier)
Per-job billing	TheirStack (1 credit per job returned)

The examples above are the limits discovered and documented during implementation (`Final_Project_Report_v2.md`, Section 7); not all 46 providers’ rate limits were independently re-verified for this report.

16. RESPONSE TIME & RELIABILITY

Provider	Duration (s)	Reliability
SmartRecruiters	385.7	High
Lever	870.1	High (slowest run despite modest volume — per-company pagination overhead)
Greenhouse	183.8	High
NHS Jobs	102.4	Undocumented endpoint (no official API contract)
USAJOBS	53.1	High
Bundesagentur für Arbeit	174.3	High (with retry logic)
Reed	56.6	High
NAV Arbeitsplassen	51.4	Rotating token (no fixed key)
Ashby	70.3	High
Workable	51.8	High

Full duration/reliability figures for all 46 providers exist in the underlying `provider_runs` table; only the ten highest-volume providers are shown here for brevity.

17. AUTHENTICATION

Auth Type	Count (of 46)
None (no auth required)	31
API key (registered)	10
Public non-secret key/token (no registration)	2
OAuth	2
HTTP Basic	1

18. FILTERING CAPABILITIES

Provider	Filtering Available
SmartRecruiters	Company, offset/limit
Greenhouse	Company only
Lever	Company, offset/limit
NHS Jobs	Keyword
USAJOBS	Page/ResultsPerPage only (no keyword/filter params implemented)
Bundesagentur für Arbeit	Keyword, location
Reed	Pagination only (no keyword/filter params implemented)
NAV Arbeitsplassen	Modified-since cursor
Ashby	Company only
Workable	Company only

Filtering capability was verified for the ten highest-volume providers; it was **not independently documented for the other 36 implemented providers** in this report. No provider was found to expose salary, experience-level, or visa-sponsorship filtering.

19. MUST HAVE / SHOULD HAVE / NICE TO HAVE

Priority	Item
Must Have	Move normalization inline into the save path
Must Have	Scheduled, automated recurring syncs
Must Have	Resolve the 5 non-contributing providers (credentials, billing, root-cause)
Should Have	Reporting dashboard on top of PostgreSQL
Should Have	Better location enrichment (~1,565 distinct unresolved country values)
Should Have	Normalized employment-type field; a department/occupation crosswalk table
Should Have	Dedicated Canada, broader APAC, and Middle East sources
Should Have	Provider-run health alerting
Should Have	Complete the 6 deferred candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)
Nice to Have	Funding-stage/company-size enrichment
Nice to Have	Salary capture where a source exposes it
Nice to Have	Historical trend tracking across multiple runs
Nice to Have	Periodic re-verification of the 268-provider research catalog

20. PRODUCTION RECOMMENDATIONS

Summarized here; full detail in Section 6. Before production reliance: close the 5-provider contribution gap, add provider-run health alerting, move normalization inline, and put the pipeline on a recurring schedule.

6. Final Recommendations

Current readiness. The platform is functionally complete for a first release: 46 providers integrated, 184,506 deduplicated jobs stored in PostgreSQL, offline country/work-arrangement normalization applied, and full fault isolation so one broken provider cannot take down a run. It reflects **one complete backfill run**, not an ongoing sync.

Remaining work, in priority order:

1. Provision missing credentials (Trade Me, CareerOneStop), resolve TheirStack's billing block, verify Fantastic.jobs' trial/account status, and re-run Arbeitnow to confirm its 0-job result was transient.
2. Pursue the two highest-value deferred candidate providers — Saramin and VDAB — both real APIs blocked only by a registration/onboarding step.
3. Add provider-run health alerting before scheduling recurring syncs, so a source silently dropping to 0 doesn't go unnoticed.
4. Move country/work-arrangement normalization inline into the ingestion path, replacing the current manual migration script.
5. Investigate the two low-yield sources (HK Gov Vacancies, MyCareersFuture) before relying on either for APAC coverage.

Production readiness. Not yet production-scheduled. Recommended before relying on this pipeline for ongoing reporting: (a) resolve the 5-provider gap above, (b) add health alerting, (c) put the pipeline on a recurring cron/schedule, (d) confirm secrets handling carries through to the deployment target, (e) extend automated test coverage from country normalization to the full aggregation/dedup pipeline.

Next steps. Treat the items above as a scoped, well-understood backlog rather than unknowns — none of them block the current dataset from being useful today. The full technical detail, evidence, and provider-by-provider reasoning behind every recommendation is in `Final_Project_Report_v2.md`.

Job Aggregation Platform — Final Project Report · Data snapshot 2026-07-10 · Report date July 15, 2026 · Condensed from `Final_Project_Report_v2.md` for a single 15–20 minute engineering-manager review. No technical finding was removed; long narrative sections were replaced with tables, and repeated explanations were cut. The Implemented Providers table (Section 4.1) was independently re-verified against `sources/*.py` on 2026-07-15, adding per-provider implementation limitations and expansion opportunities. Source of truth: PostgreSQL (`jobs` , `provider_runs`), this repository's source code, and `job_api.xlsx` (API Catalog, Sheet1 sheets). Where a fact could not be verified, this report states “Not publicly available” or “Not verified during this project” rather than estimating.